

DIGITAL SERVICE DELIVERY PLATFORM

API Integration Guide

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REVISION HISTORY

Version	Date	Author	Description
1.0	March 2026	SDP Team	Initial release

1 INTRODUCTION

The Digital Service Delivery Platform (DSDP) is the centralized integration layer that enables Content Partners (CPs) to deliver value-added services to mobile subscribers. It provides a uniform REST API surface for service activation, deactivation, SMS delivery, and real-time lifecycle notifications.

1.1 Scope

This document is intended for Content Partners and their technical teams integrating against the DSDP. It covers all supported REST APIs including authentication, subscription management, SMS delivery, bulk messaging, and outbound CP notifications.

1.2 Supported APIs.

API	Purpose
Get Token	Authenticate and obtain a JWT access token and refresh token.
Refresh Token	Renew an expired access token using the refresh token.
Subscription (Activate)	Subscribe a mobile number to a service offer.
Subscription (Deactivate)	Unsubscribe a mobile number from a service offer.
SendsMS	Deliver a content SMS to a subscriber for on-demand or subscription services.
BulkSMS	Send a single message to multiple MSISDNs in one request.
CP Notification	Outbound webhook sent by DSDP to the CP callback URL on lifecycle events.
Payment	Deduct airtime from a subscriber's account. Supports static and dynamic charging with USSD push confirmation.
Consent Gateway	Web-based consent journey that redirects subscribers to a hosted consent page for service opt-in via digital channels.

1.3 Base URLs

All API requests use HTTPS on the following base URLs. The two production endpoints are load-balanced and interchangeable — CPs may use either or both concurrently.

HTTPS	Testbed	https://dsdpstg.safaricom.com
HTTPS	Production	https://dsdp.safaricom.com

2 AUTHENTICATION

DSDP uses JWT-based Bearer token authentication. All protected endpoints require an Authorization header with a valid access token. Token has 1 hour validity window. Use the Refresh Token API to extend a session without re-authenticating.

2.1 Get Token

Exchanges CP credentials for an access token and a refresh token.

POST	Testbed	<code>https://dsdpstg.safaricom.com/api/auth/login</code>
POST	Production	<code>https://dsdp.safaricom.com/api/auth/login</code>

Request Headers

Header	Value
Content-Type	application/json
X-Requested-With	XMLHttpRequest

Request Body Parameters

Field	Description	Type	Required	Example
username	CP username assigned by SDP on registration.	String	M	cp_partner01
password	CP password assigned by SDP on registration.	String	M	P@ssw0rd!

Response Body Parameters

Field	Description	Type	Required	Example
msg	Human-readable authentication confirmation.	String	M	You have been Authenticated...
token	JWT access token. Valid for 1 hour. Use in X-Authorization header for all protected calls.	String	M	eyJhbGci...

refreshToken	Long-lived token (valid for 2 hour) used to obtain a new access token via the Refresh Token API.	String	M	eyJhbGciOi...
---------------------	--	--------	---	---------------

Sample Request

```
POST /api/auth/login
Content-Type: application/json

{
  "username": "cp_partner01",
  "password": "P@ssw0rd!"
}
```

Sample Response

[illegible]

2.2 Refresh Token

Generates a new access token from a valid refresh token, without requiring credentials.

GET	Testbed	https://dsdpstg.safaricom.com/api/auth/RefreshToken
GET	Production	https://dsdp.safaricom.com/api/auth/RefreshToken

Request Headers

Header	Value
Content-Type	application/json
X-Requested-With	XMLHttpRequest
X-Authorization	Bearer <refreshToken>

■ **Note:** This is a GET request. Pass the refresh token in the X-Authorization header. No request body is required.

Sample Response

```
{
  "token":
"eyJhbGciOiJSUzI1NiIsInR5cCIgOiAiSldUIiwia2lkIiA6ICJjZ0JxLTJvZnpMakRpeXliV2VnWS10OXZwZ0V2RDdJmVdJb180V1YxWnZJIn0.eyJleHAiOjE5NzQ0NDgzOTAsImhhdCI6MTc3NDQ0NDc5MCwianRpIjoiYjEwNWE0YTItMmZlY2S0OZWE4LWlWMDU0TmJlRiZGY1NDJlNjY2IiwiaXNzIjoiaHR0cHM6Ly9kc2RwLWlhbS5zYWZhcmljb20uY29tL3JlYWxtcy9waC1jb3JlIiwiaXVkiJpbInBoLXBvcnRhbmCIImFjY291bnQiXSwic3ViIjoiiYjUzNmJmYzktYmVlOS00NzA5LWlRMTMTEOTk1MmE2YzEzODI1IiwidHlwIjoiQmVhcmVYIiwiaXpwIjoicGFydG5lci1hcGkiLCJzZXNzLW9uX3N0YXRRIjoiYTYVY2ZgMGEtNzYxMy00MmNlLWFiMGQtNWQ5YWwMwWEzZDM4IiwiaWYiOiIjImVbG93ZnQWt3JpZ2Z1cyI6WyI6YVkiJdLCJyZWZfb3V9hY2Nlc3MiOnsic9sXZMI0LWlZGVmYXVsdCIyb2x1cyIwaC1jb3JlIiwib2ZmbGluZV9hY2Nlc3MiLCJlbnWFFfYXV0aG9yaXphdGlvbiJdfSwicmVzb3VyY2VfYWNjZXNzIjpb7InBoLXBvcnRhbmCI6eyJyb2x1cyI6WyJlZG10X3Byb19vZmYiLCJzbXNlXlYiLCJwcm9kX2MiLCJwcm9kX2UiLCJzdXNwZ5kX3Byb19vZmZlcnMiLCJ2aWV3X3IiLCJkZWxldGVfCHJvX29mZiIsInVzZXJfYWN0aXZpdHkiLCJidW5kX2UiLCJ2aWV3X3Byb19vZmYiLCJhZGRfcHJvX29mZiIsImNhbWRFdiIsImJlbnRfYyIsIm1lc3NhZ2VfcCIIsImFwcHJvdmVfcHJvX29mZiIsImJlbnRfYyIsImNvZV9hZGQlLCJvZmZlZiIsIm9mZ19jIiwiaWYiOiIjImVbG93ZnQWt3JpZ2Z1cyI6WyJlZG10X3Byb19vZmYiLCJzbXNlXlYiLCJwcm9kX2MiLCJwcm9kX2UiLCJzdXNwZ5kX3Byb19vZmZlcnMiLCJ2aWV3X3IiLCJkZWxldGVfCHJvX29mZiIsImNhbWRFdiIsImJlbnRfYyIsIm1lc3NhZ2VfcCIIsImFwcHJvdmVfcHJvX29mZiIsImJlbnRfYyIsImNvZV9hZGQlLCJvZmZlZiIsIm9mZ19jIiwiaWYiOiIjImVbG93ZnQWt3JpZ2Z1cyI6WyJlZG10X3Byb19vZmYiLCJzbXNlXlYiLCJwcm9kX2MiLCJwcm9kX2UiLCJzdXNwZ5kX3Byb19vZmZlcnMiLCJ2aWV3X3IiLCJkZWxldGVfCHJvX29mZiIsImNhbWRFdiIsImJlbnRfYyIsIm1lc3NhZ2VfcCIIsImFwcHJvdmVfcHJvX29mZiIsImJlbnRfYyIsImNvZV9hZGQlLCJvZmZlZiIsIm9mZ19jIiwiaWYiOiIjImVbG93ZnQWt3JpZ2Z1cyI6WyJlZG10X3Byb19vZmYiLCJzbXNlXlYiLCJwcm9kX2MiLCJwcm9kX2UiLCJzdXNwZ5kX3Byb19vZmZlcnMiLCJ2aWV3X3IiLCJkZWxldGVfCHJvX29mZiIsImNhbWRFdiIsImJlbnRfYyIsIm1lc3NhZ2VfcCIIsImFwcHJvdmVfcHJvX29mZiIsImJlbnRfYyIsImNvZV9hZGQlLCJvZmZlZiIsIm9mZ19jIiwiaWYiOiIjImVbG93ZnQWt3JpZ2Z1cyI6WyJlZG10X3Byb19vZmYiLCJzbXNlXlYiLCJwcm9kX2MiLCJwcm9kX2UiLCJzdXNwZ5kX3Byb19vZmZlcnMiLCJ2aWV3X3IiLCJkZWxldGVfCHJvX29mZiIsImNhbWRFdiIsImJlbnRfYyIsIm1lc3NhZ2VfcCIIsImFwcHJvdmVfcHJvX29mZiIsImJlbnRfYyIsImNvZV9hZGQlLCJvZmZlZiIsIm9mZ19jIiwiaWYiOiIjImVbG93ZnQWt3JpZ2Z1cyI6WyJlZG10X3Byb19vZmYiLCJzbXNlXlYiLCJwcm9kX2MiLCJwcm9kX2UiLCJzdXNwZ5kX3Byb19vZmZlcnMiLCJ2aWV3X3IiLCJkZWxldGVfCHJvX29mZiIsImNhbWRFdiIsImJlbnRfYyIsIm1lc3NhZ2VfcCIIsImFwcHJvdmVfcHJvX29mZiIsImJlbnRfYyIsImNvZV9hZGQlLCJvZmZlZiIsIm9mZ19jIiwiaWYiOiIjImVbG93ZnQWt3JpZ2Z1cyI6WyJlZG10X3Byb19vZmYiLCJzbXNlXlYiLCJwcm9kX2MiLCJwcm9kX2UiLCJzdXNwZ5kX3Byb19vZmZlcnMiLCJ2aWV3X3IiLCJkZWxldGVfCHJvX29mZiIsImNhbWRFdiIsImJlbnRfYyIsIm1lc3NhZ2VfcCIIsImFwcHJvdmVfcHJvX29mZiIsImJlbnRfYyIsImNvZV9hZGQlLCJvZmZlZiIsIm9mZ19jIiwiaWYiOiIjImVbG93ZnQWt3JpZ2Z1cyI6WyJlZG10X3Byb19vZmYiLCJzbXNlXlYiLCJwcm9kX2MiLCJwcm9kX2UiLCJzdXNwZ5kX3Byb19vZmZlcnMiLCJ2aWV3X3IiLCJkZWxldGVfCHJvX29mZiIsImNhbWRFdiIsImJlbnRfYyIsIm1lc3NhZ2VfcCIIsImFwcHJvdmVfcHJvX29mZiIsImJlbnRfYyIsImNvZV9hZGQlLCJvZmZlZiIsIm9mZ19jIiwiaWYiOiIjImVbG93ZnQWt3JpZ2Z1cyI6WyJlZG10X3Byb19vZmYiLCJzbXNlXlYiLCJwcm9kX2MiLCJwcm9kX2UiLCJzdXNwZ5kX3Byb19vZmZlcnMiLCJ2aWV3X3IiLCJkZWxldGVfCHJvX29mZiIsImNhbWRFdiIsImJlbnRfYyIsIm1lc3NhZ2VfcCIIsImFwcHJvdmVfcHJvX29mZiIsImJlbnRfYyIsImNvZV9hZGQlLCJvZmZlZiIsIm9mZ19jIiwiaWYiOiIjImVbG93ZnQWt3JpZ2Z1cyI6WyJlZG10X3Byb19vZmYiLCJzbXNlXlYiLCJwcm9kX2MiLCJwcm9kX2UiLCJzdXNwZ5kX3Byb19vZmZlcnMiLCJ2aWV3X3IiLCJkZWxldGVfCHJvX29mZiIsImNhbWRFdiIsImJlbnRfYyIsIm1lc3NhZ2VfcCIIsImFwcHJvdmVfcHJvX29mZiIsImJlbnRfYyIsImNvZV9hZGQlLCJvZmZlZiIsIm9mZ19jIiwiaWYiOiIjImVbG93ZnQWt3JpZ2Z1cyI6WyJlZG10X3Byb19vZmYiLCJzbXNlXlYiLCJwcm9kX2MiLCJwcm9kX2UiLCJzdXNwZ5kX3Byb19vZmZlcnMiLCJ2aWV3X3IiLCJkZWxldGVfCHJvX29mZiIsImNhbWRFdiIsImJlbnRfYyIsIm1lc3NhZ2VfcCIIsImFwcHJvdmVfcHJvX29mZiIsImJlbnRfYyIsImNvZV9hZGQlLCJvZmZlZiIsIm9mZ19jIiwiaWYiOiIjImVbG93ZnQWt3JpZ2Z1cyI6WyJlZG10X3Byb19vZmYiLCJzbXNlXlYiLCJwcm9kX2MiLCJwcm9kX2UiLCJzdXNwZ5kX3Byb19vZmZlcnMiLCJ2aWV3X3IiLCJkZWxldGVfCHJvX29mZiIsImNhbWRFdiIsImJlbnRfYyIsIm1lc3NhZ2VfcCIIsImFwcHJvdmVfcHJvX29mZiIsImJlbnRfYyIsImNvZV9hZGQlLCJvZmZlZiIsIm9mZ19jIiwiaWYiOiIjImVbG93ZnQWt3JpZ2Z1cyI6WyJlZG10X3Byb19vZmYiLCJzbXNlXlYiLCJwcm9kX2MiLCJwcm9kX2UiLCJzdXNwZ5kX3Byb19vZmZlcnMiLCJ2aWV3X3IiLCJkZWxldGVfCHJvX29mZiIsImNhbWRFdiIsImJlbnRfYyIsIm1lc3NhZ2VfcCIIsImFwcHJvdmVfcHJvX29mZiIsImJlbnRfYyIsImNvZV9hZGQlLCJvZmZlZiIsIm9mZ19jIiwiaWYiOiIjImVbG93ZnQWt3JpZ2Z1cyI6WyJlZG10X3Byb19vZmYiLCJzbXNlXlYiLCJwcm9kX2MiLCJwcm9kX2UiLCJzdXNwZ5kX3Byb19vZmZlcnMiLCJ2aWV3X3IiLCJkZWxldGVfCHJvX29mZiIsImNhbWRFdiIsImJlbnRfYyIsIm1lc3NhZ2VfcCIIsImFwcHJvdmVfcHJvX29mZiIsImJlbnRfYyIsImNvZV9hZGQlLCJvZmZlZiIsIm9mZ19jIiwiaWYiOiIjImVbG93ZnQWt3JpZ2Z1cyI6WyJlZG10X3Byb19vZmYiLCJzbXNlXlYiLCJwcm9kX2MiLCJwcm9kX2UiLCJzdXNwZ5kX3Byb19vZmZlcnMiLCJ2aWV3X3IiLCJkZWxldGVfCHJvX29mZiIsImNhbWRFdiIsImJlbnRfYyIsIm1lc3NhZ2VfcCIIsImFwcHJvdmVfcHJvX29mZiIsImJlbnRfYyIsImNvZV9hZGQlLCJvZmZlZiIsIm9mZ19jIiwiaWYiOiIjImVbG93ZnQWt3JpZ2Z1cyI6WyJlZG10X3Byb19vZmYiLCJzbXNlXlYiLCJwcm9kX2MiLCJwcm9kX2UiLCJzdXNwZ5kX3Byb19vZmZlcnMiLCJ2aWV3X3IiLCJkZWxldGVfCHJvX29mZiIsImNhbWRFdiIsImJlbnRfYyIsIm1lc3NhZ2VfcCIIsImFwcHJvdmVfcHJvX29mZiIsImJlbnRfYyIsImNvZV9hZGQlLCJvZmZlZiIsIm9mZ19jIiwiaWYiOiIjImVbG93ZnQWt3JpZ2Z1cyI6WyJlZG10X3Byb19vZmYiLCJzbXNlXlYiLCJwcm9kX2MiLCJwcm9kX2UiLCJzdXNwZ5kX3Byb19vZmZlcnMiLCJ2aWV3X3IiLCJkZWxldGVfCHJvX29mZiIsImNhbWRFdiIsImJlbnRfYyIsIm1lc
```



```
D2brtzdKpx5SdC1z_lrvGIgisEmfJeVjDglyixwGgxarQcE7NB4fMg6xs0BA3apQGuHdR0upj9n7ZgbulVuewXoO-  
y6LXUjkxUc2_Ojq_PFy8BN_tMUV7xeEog"  
}
```

3 SUBSCRIPTION MANAGEMENT

3.1 Activate Subscription

Subscribes a mobile number to a service offer. Returns a synchronous response indicating whether the activation was accepted.

POST	Testbed	<code>https://dsdpstg.safaricom.com/api/public/SDP/activate</code>
POST	Production	<code>https://dsdp.safaricom.com/api/public/SDP/activate</code>

Request Headers

Header	Value
Content-Type	application/json
X-Requested-With	XMLHttpRequest
X-Authorization	Bearer <accessToken>

Request Body Parameters

Field	Description	Type	Required	Example
<code>requestId</code>	Unique client transaction ID for tracking. Maximum 50 characters.	String	M	60141234567890
<code>channel</code>	Operation channel initiating the request. Valid channel values are shared during integration.	String	M	APIGW
<code>operation</code>	API operation type. Set to ACTIVATE for this endpoint.	String	M	ACTIVATE
<code>OfferCode</code>	Service offer code allocated by Safaricom or the partner upon service creation.	String	M	350032100559
<code>MSISDN</code>	Subscriber mobile number value of this field is mobile number or MDN without country code or '0' in the beginning. . For hashed-enabled services, pass the encrypted MSISDN.	String	M	Plain: 254720123456 Encrypted: 904586536756
<code>CpId</code>	Partner ID allocated at registration.	String	M	EXT_ID_10

Sample Request

```
POST /api/public/SDP/activate
Content-Type: application/json
X-Authorization: Bearer eyJhbGci...

{
  "requestId": 17,
  "channel": "APIGW",
  "operation": "ACTIVATE",
  "requestParam": {
    "data": [
      { "name": "OfferCode", "value": "350032100559" },
      { "name": "Msisdn", "value": "254795421629" },
      { "name": "Language", "value": "1" },
      { "name": "CpId", "value": "EXT_ID_10" }
    ]
  }
}
```

Response Parameters

Field	Description	Type	Required	Example
requestId	Echoed unique transaction ID from the request.	String	M	60141234567890
responseTimeStamp	Response timestamp. Format: yyyyMMddHHmmss.	String	M	20260326142215
channel	Operation channel.	String	M	APIGW
operation	Operation type from the request.	String	M	ACTIVATE
ResponseId	SDP-generated transaction ID for tracking.	String	M	50120155555
status	Result flag. 0 = success, 1 = failure.	String	M	0
statusCode	Numeric status code. See Error Code Reference.	String	M	0
description	Human-readable status description.	String	M	Activation successful

Sample Response

```
{
  "requestId": "17",
  "responseId": "cp2910183038077087336761",
  "responseTimeStamp": "20260326092806",
  "channel": "APIGW",
  "operation": "ACTIVATE",
  "requestParam": {
    "data": [
      { "name": "OfferCode", "value": "350032100559" },
      { "name": "Msisdn", "value": "254795421629" },
      { "name": "Language", "value": "1" },
      { "name": "CpId", "value": "EXT_ID_10" }
    ]
  },
  "responseParam": {
    "status": "0",
    "statusCode": "112",
    "description": "Thank you customer, Your Request for the Service 21234_testoffer_ksh0_PerWeek is under process."
  }
}
```

3.2 Deactivate Subscription

Unsubscribes a mobile number from a service offer. The operation field must be set to DEACTIVATE.

POST	Testbed	https://dsdpstg.safaricom.com/api/public/SDP/deactivate
POST	Production	https://dsdp.safaricom.com/api/public/SDP/deactivate

Request Headers

Header	Value
Content-Type	application/json
X-Requested-With	XMLHttpRequest
X-Authorization	Bearer <accessToken>

Request Body Parameters

Field	Description	Type	Required	Example
requestId	Unique client transaction ID for tracking. Maximum 50 characters.	String	M	60141234567890
channel	Operation channel. Valid values shared during integration.	String	M	APIGW
operation	Must be set to DEACTIVATE.	String	M	DEACTIVATE
OfferCode	Service offer code.	String	M	1001

MSISDN	Subscriber mobile number value of this field is mobile number or MDN without country code or '0' in the beginning. . For hashed-enabled services, pass the encrypted MSISDN.	String	M	720123456
CpId	Partner ID allocated at registration.	String	M	EXT_ID_10

Sample Request

```
POST /api/public/SDP/deactivate
Content-Type: application/json
X-Authorization: Bearer eyJhbGci...

{
  "requestId": 17,
  "channel": "APIGW",
  "operation": "DEACTIVATE",
  "requestParam": {
    "data": [
      { "name": "OfferCode", "value": "1001" },
      { "name": "Msisdn", "value": "254720123456" },
      { "name": "CpId", "value": "EXT_ID_10" }
    ]
  }
}
```

Sample Response

```
{
  "requestId": "17",
  "responseId": "cp2910183038077087336761",
  "responseTimeStamp": "20260326092806",
  "channel": "APIGW",
  "operation": "ACTIVATE",
  "requestParam": {
    "data": [
      { "name": "OfferCode", "value": "1001" },
      { "name": "Msisdn", "value": "254720123456" },
      { "name": "CpId", "value": "EXT_ID_10" }
    ]
  },
  "responseParam": {
    "status": "0",
    "statusCode": "1",
    "description": "Deactivate subscription success"
  }
}
```

4 SMS DELIVERY

4.1 SendSMS

Sends a single SMS to a subscriber. Used for both on-demand and subscription-based content delivery. For on-demand services, the LinkId parameter is mandatory to associate the message with the original MO request.

POST	Testbed	<code>https://dsdpstg.safaricom.com/api/public/SDP/sendSMSRequest</code>
POST	Production	<code>https://dsdp.safaricom.com/api/public/SDP/sendSMSRequest</code>

Request Headers

Header	Value
Content-Type	application/json
X-Requested-With	XMLHttpRequest
X-Authorization	Bearer <accessToken>

Request Body Parameters

Field	Description	Type	Required	Example
requestId	Unique client transaction ID. Maximum 50 characters.	String	M	60141234567890
channel	Operation channel. Valid values shared during integration.	String	M	APIGW
operation	Must be set to SendSMS.	String	M	SendSMS
OfferCode	Service offer code.	String	M	5012015555
MSISDN	Subscriber mobile number value of this field is mobile number or MDN without country code or '0' in the beginning. . For hashed-enabled services, pass the encrypted MSISDN.	String	M	Plain: 254720123456 Encrypted: 904586536756
CpId	Partner ID allocated at registration.	String	M	EXT_ID_10
LinkId	Session ID generated when the subscriber initiated an MO	String	M (on-demand)	00010310189519161781865526

	request. Mandatory for on-demand services.			
Content	SMS message body to be delivered to the subscriber.	String	M	Thank you for subscribing.

Sample Request

```
POST /api/public/SDP/sendSMSRequest
Content-Type: application/json
X-Authorization: Bearer eyJhbGci...
```

```
{
  "requestId": 17,
  "channel": "APIGW",
  "operation": "SendSMS",
  "requestParam": {
    "data": [
      { "name": "LinkId", "value": "00010310189519161781865526" },
      { "name": "Msisdn", "value": "254795421629" },
      { "name": "Content", "value": "Your daily content is ready. Enjoy!" },
      { "name": "OfferCode", "value": "1003" },
      { "name": "CpId", "value": "EXT_ID_10" }
    ]
  }
}
```

Sample Response

```
{
  "requestId": "17",
  "responseId": "cp2910183038077087336761",
  "responseTimeStamp": "20260326092806",
  "channel": "APIGW",
  "operation": "ACTIVATE",
  "requestParam": {
    "data": [
      { "name": "LinkId", "value": "00010310189519161781865526" },
      { "name": "Msisdn", "value": "254795421629" },
      { "name": "Content", "value": "Your daily content is ready. Enjoy!" },
      { "name": "OfferCode", "value": "1003" },
      { "name": "CpId", "value": "EXT_ID_10" }
    ]
  },
  "responseParam": {
    "status": "0",
    "statusCode": "849",
    "description": " Thank you customer, Your Request for the Service 21234_testoffer_ksh0_PerWeek is under process."
  }
}
```

4.2 BulkSMS

Sends a single message to one or more MSISDNs in a single API call. The platform returns a synchronous acceptance response. Delivery status notifications are sent asynchronously to the `actionResponseURL` if provided.

POST	Testbed	https://dsdpstg.safaricom.com/api/public/CMS/bulksms
POST	Production	https://dsdp.safaricom.com/api/public/CMS/bulksms

Request Body Parameters

Field	Description	Type	Required	Example
<code>timeStamp</code>	Unix epoch timestamp in milliseconds at request time.	Long	M	15567981561235
<code>userName</code>	Partner ID allocated at registration.	String	M	EXT_ID_10
<code>channel</code>	Message channel. Typically sms.	String	M	sms
<code>oa</code>	Originating address (sender name) assigned to the partner.	String	M	TestSender
<code>hashed</code>	Indicates MSISDN type. yes = encrypted MSISDNs, no = plain MSISDNs.	String	Conditional	yes
<code>msisdn</code>	Comma-separated list of destination mobile numbers.	String	M	254758934575,254795421629
<code>message</code>	SMS content to be delivered.	String	M	Hello from our service!
<code>uniqueId</code>	Alphanumeric request identifier, echoed back in responses.	String	M	2500688298721
<code>actionResponseURL</code>	Callback URL for asynchronous delivery receipts.	String	O	https://your.server/dlr
<code>packageId</code>	Bulk SMS package ID. Omit to allow automatic package selection.	Integer	O	4391

Sample Request

```
POST /api/public/CMS/bulksms
Content-Type: application/json
X-Authorization: Bearer eyJhbGci...

{
  "timeStamp": 15567981561235,
  "dataSet": [{
    "userName": "cpon28user",
    "channel": "sms",
    "oa": "MySender",
    "hashed": "no",
    "msisdn": "254758934575,254795421629",
    "message": "Hello! Your weekly update is ready.",
    "uniqueId": "2500688298721",
    "actionResponseURL": "https://www.example.com/dlr"
  }]
}
```

Sample Response

```
{
  "keyword": "Bulk",
  "status": "SUCCESS",
  "statusCode": "SC0000"
}
```

5 PAYMENT

The Payment API allows third parties to deduct airtime from a subscriber's account. When triggered, a USSD push may be sent to the subscriber for confirmation, after which DSDP initiates charging against the subscriber's number. SDP uses the callback configured on the service to notify the CP whether charging was successful. A callback is only sent if the synchronous response was a success.

DSDP supports both fixed and dynamic charging. Fixed charging is recommended: a fixed amount is configured on the offer and the CP must pass that exact amount in the request. With dynamic charging an amount range is configured and the CP can specify any amount within that range per day per MSISDN. If a subscriber has already been successfully charged for the day and the same offer is invoked again, DSDP returns an error.

5.1 Payment Request

POST	Testbed	https://dsdpstg.safaricom.com/api/public/SDP/paymentRequest
POST	Production	https://dsdp.safaricom.com/api/public/SDP/paymentRequest

Request Headers

Field	Description	Type	Required	Example
Content-Type	Must be application/json	String	M	application/json
X-Requested-With	Must be XMLHttpRequest	String	M	XMLHttpRequest
X-Authorization	Bearer access token	String	M	Bearer eyJhbGci...

Request Body Parameters

Field	Description	Type	Required	Example
requestId	Unique client tracking ID. Format: CPName + 16-digit random number.	String	M	MyCP1234567890123456
channel	Operation channel. Set to APIGW.	String	M	APIGW
operation	Must be set to PAYMENT.	String	M	PAYMENT
OfferCode	Service offer code (format: SPID+CPID+5digit).	String	M	350032100559
Msisdn	Subscriber mobile number in format 254xxxxxxxx or 7xxxxxxx (without country code or leading 0). For hashed-enabled services, pass the encrypted MSISDN.	String	M	254720123456
CpId	Partner ID allocated at registration.	String	M	EXT_ID_10

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ChargeAmount	Amount to be charged. Must match the configured static amount or fall within the dynamic range.	String	M	10
--------------	---	--------	---	----

Sample Request

POST /api/public/SDP/paymentRequest

Content-Type: application/json

X-Authorization: Bearer eyJhbGci...

```
{
  "requestId": "MyCP1234567890123456",
  "channel": "APIGW",
  "operation": "PAYMENT",
  "requestParam": {
    "data": [
      { "name": "OfferCode", "value": "350032100559" },
      { "name": "Msisdn", "value": "254720123456" },
      { "name": "CpId", "value": "EXT_ID_10" },
      { "name": "ChargeAmount", "value": "10" }
    ]
  }
}
```

Response Parameters

Field	Description	Type	Required	Example
requestId	Echoed client tracking ID from the request.	String	M	MyCP1234567890123456
responseTimeStamp	Response timestamp. Format: yyyy-MM-dd HH:mm:ss.SSS	String	M	2026-03-26 09:28:06.000
channel	Operation channel.	String	M	APIGW
operation	Always PAYMENT.	String	M	PAYMENT
responseId	SDP-generated transaction ID.	String	M	cp2910183038077087336761
status	0 = success, 1 = failure.	String	M	0
statusCode	Numeric code. 805 = Payment under process, 804 = Quota exceeded, 828 = Subscriber not found, 4012 = Insufficient funds.	String	M	805
description	Human-readable status description.	String	M	Payment under process
ChargeAmount	Amount charged to the subscriber.	String	M	10

Sample Response

HTTP 200 OK

```
{
  "requestId": "MyCP1234567890123456",
  "responseId": "cp2910183038077087336761",
```

```
"responseTimeStamp": "2026-03-26 09:28:06.000",
"channel": "APIGW",
"operation": "PAYMENT",
"requestParam": {
  "data": [
    { "name": "OfferCode",      "value": "350032100559" },
    { "name": "Msisdn",        "value": "254720123456" },
    { "name": "CpId",          "value": "EXT_ID_10" },
    { "name": "ChargeAmount",  "value": "10" }
  ]
},
"responseParam": {
  "status": "0",
  "statusCode": "852",
  "description": "Payment under process",
}
}
```

6 CONSENT GATEWAY

The Consent Gateway provides a hosted web-based consent journey for services that use a digital web channel. Instead of triggering a USSD consent prompt, the Content Partner redirects the subscriber to a Safaricom-hosted consent page where they can explicitly opt in to the service. This is the recommended approach for services integrated via web or app digital journeys.

6.1 How It Works

1. The CP constructs the Consent Gateway URL using the service OfferCode and redirects the subscriber's browser to it.
2. The subscriber reviews the service terms on the hosted Safaricom consent page and accepts or declines.

6.2 Consent Gateway URL

Redirect the subscriber's browser to the following URL, substituting the service OfferCode:

Staging:

Consent Gateway URL Format

```
https://dsdpstg-cg.safaricom.com/consent-gateway/{OfferCode}
```

Example (OfferCode = 300001507):

```
https://dsdp-cg.safaricom.com/consent-gateway/300000000
```

Staging:

Consent Gateway URL Format

```
https://dsdp-cg.safaricom.com/consent-gateway/{OfferCode}
```

Example (OfferCode = 300001507):

```
https://dsdp-cg.safaricom.com/consent-gateway/300000000
```

Field	Description	Type	Required	Example
OfferCode	The service offer code allocated by Safaricom. Replace {OfferCode} in the URL with the actual code.	String	M	300001507

7 CP NOTIFICATIONS

CP Notifications are outbound webhooks that DSDP sends to the Content Partner's registered callback URL to report subscriber lifecycle events. CPs must respond with an HTTP 200 and the acknowledgement payload below.

5.1 Acknowledgement Response

Respond to every CP Notification with the following JSON payload:

```
{
  "requestId": "<echo requestId from notification>",
  "responseId": "<SDP responseId>",
  "responseTimeStamp": "<yyyyMMddHHmmss>",
  "operation": "CP_NOTIFICATION",
  "responseParam": {
    "status": "0",
    "statusCode": "0000",
    "description": "Success"
  }
}
```

7.2 Notification Parameters

Field	Description	Type	Required	Example
requestId	SDP-generated notification request ID.	String	M	17223344555
requestTimeStamp	Timestamp of the notification. Format: yyyyMMddHHmmss.	String	M	20260326145324
operation	Always CP_NOTIFICATION for standard events.	String	M	CP_NOTIFICATION
MSISDN	Mobile number of the subscriber.	String	M	254716848648
Command	Event type — see Command Types table below.	String	M	ACTIVATION
OfferCode	Service offer code. Included for most command types.	String	C	1001
SubscriptionStatus	Current subscription status: A=Active, G=Grace, S=Suspend, D=Deactivated.	String	O	A
SubscriberLifeCycle	Lifecycle stage code. E.g. SUB1=Activation, REN1=Renewal, UNSUB1=Deactivation.	String	C	SUB1

TransactionId	SDP transaction reference. Not sent for delivery receipts or interactive responses.	String	C	10188826109958294032
ClientTransactionId	Original client transaction ID if provided in the triggering request.	String	O	12
ChargeAmount	Amount charged during activation, renewal, payment or content charging. Zero for deactivations.	String	C	5.00
BillingId	Billing reference for refund operations. Null for deactivations.	String	C	BID12345
NextBillingDate	Next billing date for active subscriptions. Format: yyyyMMddHHmmss.	String	C	20260416145324
Reason	Deactivation reason: USER_INITIATED or SYSTEM_INITIATED.	String	C	USER_INITIATED
Channel	Channel on which the original request was initiated.	String	O	10
Language	Subscriber preferred language code.	String	O	1

5.3 Command Types

Command	Description
ACTIVATION	Subscriber has successfully activated a service.
DEACTIVATION	Subscriber has unsubscribed from a service.
RENEWAL	Subscription has been renewed (billing cycle).
PAYMENT	A one-off payment transaction has completed.
CONTENT_CHARGE	Content charge has been applied to the subscriber.
REFUND	A refund has been processed for the subscriber.
NOTIFY_LINKID	Inbound MO request from a subscriber for on-demand service.
DELIVER_RECEIPT (Content)	Delivery status of a content SMS sent via SendSMS.
DELIVER_RECEIPT (Bulk)	Delivery status of a bulk message.
INTERACTIVE_RESPONSE	MO SMS received from a subscriber for an interactive bulk service.
CONSENT	Subscriber has responded to a USSD consent prompt.
CP_BLACKLIST	CP has been blacklisted by the platform.

5.4 Notification Examples

ACTIVATION

Sent when a subscriber successfully activates a service offer.

```
{
  "requestId": "12",
  "requestTimeStamp": "20260326091747",
  "requestParam": {
    "data": [
      { "name": "OfferCode",          "value": "001006300035" },
      { "name": "TransactionId",      "value": "10188826109958294032" },
      { "name": "ClientTransactionId", "value": "12" },
      { "name": "Language",           "value": "1" },
      { "name": "SubscriberLifeCycle", "value": "Subscription" },
      { "name": "SubscriptionStatus",  "value": "A" },
      { "name": "Channel",             "value": "10" },
      { "name": "Type",                "value": "ACTIVATION" },
      { "name": "OfferName",           "value": "9200_Daily_ksh1" },
      { "name": "Msisdn",              "value": "254728171730" }
    ]
  },
  "operation": "CP_NOTIFICATION"
}
```

DEACTIVATION

Sent when a subscriber cancels or is unsubscribed from a service offer.

```
{
  "requestId": "12",
  "requestTimeStamp": "20260326091747",
  "requestParam": {
    "data": [
      { "name": "OfferCode",          "value": "001006300035" },
      { "name": "TransactionId",      "value": "10188826109958294032" },
      { "name": "SubscriberLifeCycle", "value": "Unsubscription" },
      { "name": "SubscriptionStatus",  "value": "D" },
      { "name": "Reason",              "value": "USER_INIT_REQUEST" },
      { "name": "Type",                "value": "DEACTIVATION" },
      { "name": "OfferName",           "value": "9200_Daily_ksh1" },
      { "name": "Msisdn",              "value": "254728171730" }
    ]
  },
  "operation": "CP_NOTIFICATION"
}
```


NOTIFY_LINKID

Sent for on-demand services when a subscriber sends an MO message. The LinkId must be included in the subsequent SendSMS response.

```
{
  "requestId": "39856",
  "requestTimeStamp": "20260326185725",
  "requestParam": {
    "data": [
      { "name": "LinkId", "value": "0007311018784515661497991" },
      { "name": "OfferCode", "value": "001007300127" },
      { "name": "RefernceId", "value": "1018784515649514444" },
      { "name": "ClientTransactionId", "value": "39856" },
      { "name": "Type", "value": "NOTIFY_LINKID" },
      { "name": "USER_DATA", "value": "INFO" },
      { "name": "Msisdn", "value": "254748250794" }
    ]
  },
  "operation": "CP_NOTIFICATION"
}
```

DELIVERY_RECEIPT

Sent after a SendSMS request to report the final delivery status of the message.

```
{
  "requestId": "10181687073878464681",
  "requestTimeStamp": "20260326112250",
  "requestParam": {
    "data": [
      { "name": "Refund", "value": "0" },
      { "name": "Type", "value": "DELIVERY_RECEIPT" },
      { "name": "Description", "value": "DeliveryToTerminal" },
      { "name": "ClientTransactionId", "value": "880630501079943782" },
      { "name": "Msisdn", "value": "254728171730" }
    ]
  },
  "operation": "CP_NOTIFICATION"
}
```

CONSENT

Sent when a subscriber responds to a USSD consent prompt. The ConsentValue field contains the subscriber's choice.

```
{
  "requestId": "10185548875490033023",
  "requestTimeStamp": "20260326222328",
  "requestParam": {
    "data": [
      { "name": "OfferCode", "value": "001011900448" },
      { "name": "Language", "value": "1" },
      { "name": "CpId", "value": "119" },
      { "name": "Type", "value": "CONSENT" },
      { "name": "Msisdn", "value": "254748249894" }
    ],
    "additionalData": [
      { "name": "ConsentValue", "value": "Yes" }
    ]
  },
  "operation": "CP_NOTIFICATION"
}
```

8 ERROR CODE REFERENCE

6.1 Subscription & On-Demand Error Codes

Error Code	Description	Status	Status Code
0	Activation Success	Success	0
1	Deactivation Success	Success	0
148	Promotional Activation Success	Success	0
327	Subscription — Special Number Applied	Success	0
780	On-demand Request Success	Success	0
781	Request Under Process	Success	0
5	Subscription Does Not Exist	Failure	1
6	Subscription Already Expired	Failure	1
9	Subscription Already Exists	Failure	1
10	Offer Expired	Failure	1
11	Offer Does Not Exist	Failure	1
15	Deactivation Failure	Failure	1
17	Subscription Already Deactivated	Failure	1
19	Offer Is Not Active	Failure	1
41	Subscriber Number Invalid	Failure	1
102	Mandatory Parameter Missing	Failure	1
123	Consent Prompt Sent — Awaiting Reply	Success	0
288	Internal Error	Failure	1
782	Charging Failure: Insufficient Funds	Failure	1
826	Shortcode Blacklisted	Failure	1
827	Service in Freeze Period	Failure	1
849	Send SMS In Progress	Success	0
858	LinkId Expired or Does Not Exist	Failure	1

8.2 Payment Error Codes

Error Code	Description	Status Code
852	Payment Charging Success	0
804	Payment Quota Exceeded	1

828	Subscriber Profile Not Found	1
-----	------------------------------	---

8.3 Consent Error Codes

Error Code	Description	Status Code
121	Subscriber Declined Consent	1
293	Consent Reply Pending	0
317	Consent Message Send Failure	1
816	Subscriber Cancelled Consent	1

8.4 BulkSMS Error Codes

Error Code	Description	Process Type
SC0000	Success	Sync / Async
SC0001	User ID Missing	Async
SC0002	Critical Error	Async
SC0003	Campaign ID Missing	Async
SC0004	Username Missing	Async
SC0005	Plan ID Missing	Async
SC0006	Incorrect Username	Async
SC0007	User Blocked / Validation Failed	Async
SC0008	Failure / Mandatory Param Missing	Async
SC0009	Quota Threshold Reached	Async
SC0010	Database Insertion Failed	Async
SC0011	Total Quota Exceeded	Async
SC0012	Quota Expired	Async
SC0036	Hashed Parameter Missing	Sync
SC0036	Invalid Hashed Tag Sent	Sync
SC0039	Only Clear MSISDN Allowed	Sync
SC0040	Only Hashed MSISDN Allowed	Sync
SC0042	Missing Value For Hashed Tag	Sync

6.5 SMPP Delivery Error Codes

SMPP Error Code	Description
0	Delivered to Terminal — OK
1	Delivery Impossible
5	Delivery Impossible
6	Absent Subscriber
11	Delivery Impossible
13	Delivery Impossible
27	Absent Subscriber
31	Delivery Impossible
32	Delivery Impossible
34	Delivery Impossible
99	Delivery Impossible
1001	Network Failure
1020	Sendername Blacklisted
2020	Invalid MSISDN